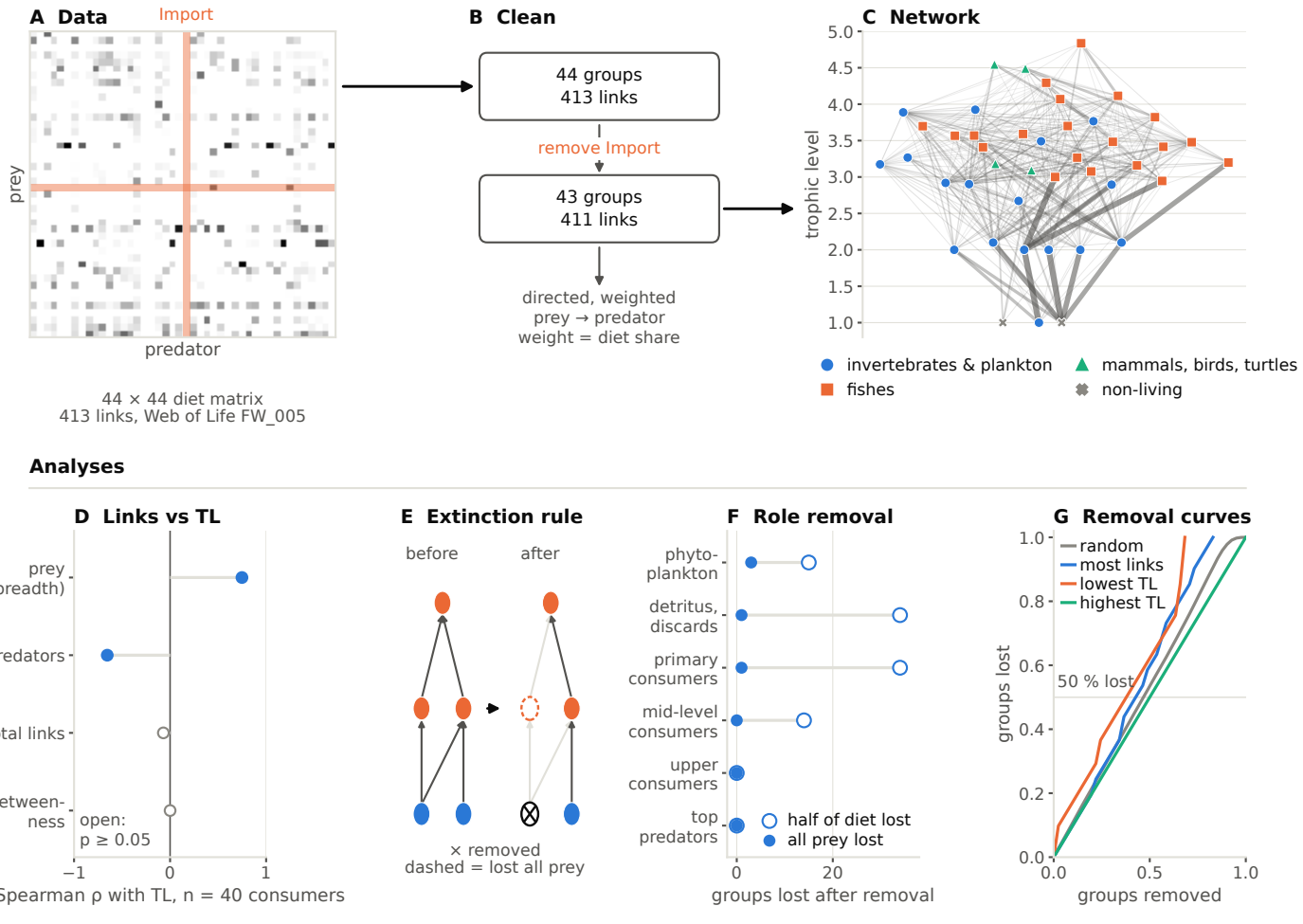


(A) The raw Gulf of Cadiz diet matrix (44×44), with columns representing predators and rows representing prey. Darker cells indicate that a prey group makes up a larger proportion of a predator's diet. The "Import" row and column represent food sources originating outside the Gulf of Cadiz. **(B)** The processed diet matrix after removing the "Import" row and column, leaving 43 functional groups and 411 feeding links. **(C)** Network visualization of the processed food web, with each node representing a functional group and each edge representing a feeding relationship. Edge thickness indicates the proportion of a predator's diet contributed by its prey. Nodes are arranged vertically by trophic level, with primary producers at the bottom and top predators near the top. **(D)** Each row ranks the 40 consumer groups by trophic level and by one count of connections, and the dot shows how closely the two rankings match. The three basal groups are left out because they have no prey by definition. Groups higher in the web eat more kinds of prey and have fewer predators, but these cancel out in total links and betweenness, showing no clear trend. **(E)** Diagram showing the extinction rule: a consumer group will die out when it loses its prey. **(F)** Effects of removing entire trophic roles from the food web. Each row represents the removal of a trophic role, and the points indicate the number of additional groups that go extinct as a result. Filled points represent groups that lose all their prey, while open points represent groups that lose at least half of their prey. Removing lower trophic roles triggers extinction cascades, whereas removing upper trophic roles causes few or no additional extinctions. **(G)** Each line removes groups one at a time, applies the extinction rule so that groups with no prey left also die, and plots the share of the web lost against the share removed, so curves above the dashed diagonal mean cascades are happening. Removing the lowest trophic levels first does the most damage, while removing the highest first causes no extra losses.